mergencystretch

Chapter 1. Introduction and Causal Models

Pearl's causal hierarchy, Simpson's paradox, structural causal models, causal graphs, d-separation

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1.1 Why causal inference

Motivation. The central question is why the tools of classical machine learning and statistics, which learn aspects of a *fixed* data-generating reality, are not sufficient to answer questions about actions and about worlds that were never realized.

1.1.1 A brief review of machine learning

Motivation. Describe what standard learning paradigms estimate, so that it becomes clear what they *cannot* estimate.

1.1.1.1 Why machine learning?

AI/ML is the design and analysis of intelligent systems. Intelligent behavior requires knowledge, i.e. a model of the environment. Writing this knowledge down explicitly is hard and often infeasible, so it is acquired by *learning from data through algorithms*.

Learning modifies an agent's decision mechanism in order to improve performance. Since environments change, an agent must adapt to unexpected situations; learning is essential when the designer is not omniscient, and is particularly useful for messy real-world data. Three fundamental questions recur:

- How can the learning process be characterized?
- How should prior knowledge be traded off against performance?
- How robust and generalizable is the resulting system?

Applications are everywhere: generative AI (LLMs, multimodal agents, digital twins), AI for science (protein folding, drug discovery, materials), medicine (personalized treatment, surgical planning), sustainability (climate, weather, energy grids), vision (scene understanding, deepfake detection), autonomous systems (humanoid robots, drone fleets) and security (fraud detection, cyber-security agents).

1.1.1.2 The classical learning paradigm

There is an unknown reality, a true model M, that generates the world. M produces examples represented as feature vectors x, and one is interested in an ideal target function $f_M(x)$ of that reality. From training data $\{x^{(i)}\}_{i=1}^N$ one selects a hypothesis $h \in \mathcal{H}$ such that $h(x) \approx f_M(x)$ for every instance. The main challenge is generalization to unseen instances.

Supervised learning maps a feature space $\mathcal{X}$ to a label space $\mathcal{Y}$ from labeled pairs $\{(x^{(i)}, y^{(i)})\}_{i=1}^N$, looking for h that mimics an oracle $f : \mathcal{X} \to \mathcal{Y}$. Common hypothesis classes are non-parametric methods (decision trees, k-NN), probabilistic models (Bayesian networks), deep networks (CNNs, Transformers) and ensembles (XGBoost, LightGBM).

Unsupervised learning receives only $x \in \mathcal{X}$ and seeks a representation revealing intrinsic structure: clustering, dimensionality reduction (latent spaces), and generative modeling of the density $P(x)$. (A phylogenetic tree, a hierarchical clustering of species, is a typical example.)

1.1.1.3 A probabilistic view

Both paradigms estimate features of one fixed joint distribution:

- supervised learning is essentially inference of the conditional distribution $P(y \mid x)$ (e.g. the output activation of a network or a leaf of a decision tree);
- unsupervised learning infers properties of the marginal distribution $P(x)$.

In both cases *we learn to imitate aspects of an underlying static reality.* Nothing in $P(x, y)$ tells us what would happen if the mechanism producing x were changed.

1.1.1.4 Reinforcement learning: intervention instead of observation

In reinforcement learning no static labeled sample is given. The agent *intervenes* by interacting with the environment and learns from its responses (A/B testing, robotics). The key difficulty is the exploration/exploitation trade-off. RL thus touches a new kind of quantity, the effect of an action; the course asks what is still missing from this picture, in particular how to reason about actions using data that were *not* generated by our own experiments, and how to reason about events that did not happen.

1.1.2 Conceptual foundation: the ladder of causation

Motivation. Introduce the three cognitive abilities, observing, intervening, imagining, that structure the whole course.

Human cognition can be organized in three layers:

Layer 1. Observation (sensors): the ability to observe the world and infer the "next" stage.

Layer 2. Agency: the ability to intervene in the world and change it.

Layer 3. Retrospection: the ability to reflect on what we did (and did not do) and infer new behaviors.

This cognitive stratification is formalized as **Pearl's Causal Hierarchy (PCH)**.

1.1.2.1 Inference about worlds that are not realized

The classical pipeline is

$$\text{Real world / nature} \xrightarrow{\text{data}} \text{AI/ML, statistics (inference)} \longrightarrow \text{new knowledge.}$$

For an *alternative reality* that was not realized ("what if we had changed the policy?") there are no data at all. The only route is

$$\text{knowledge} \longrightarrow \text{causal model} \xrightarrow{\text{causal inference}} \text{new knowledge about the alternative reality.}$$

A causal model carries assumptions about mechanisms that let us transport information from the observed world to unobserved ones.

1.1.2.2 Pearl's Causal Hierarchy

Level (symbol)	Activity	Question	Examples / typical models
1. Association	Seeing	What is?	What does a symptom tell us about a disease? How would seeing X change my belief in Y?
$P(y \mid x)$			(Un)supervised ML: deep nets, Bayes nets, hierarchical models, decision trees.
2. Intervention	Doing	What if?	What if I take aspirin, will my headache be cured? What if I do X?
$P(y \mid \mathrm{do}(x), c)$			Reinforcement learning, causal Bayesian networks, MDPs, POMDPs.
3. Counterfactual	Imagining,	Why?	Was it the aspirin that stopped my headache? What if I had acted differently?
$P(y_x \mid x', y')$	retrospection		Structural causal models.

Causal inference = cross-layer inference. Typical tasks take data from one layer and answer a query from a higher (or different) layer:

- *Data from Layer 1, query in Layer 2.* "Will students' performance improve if we change the class time to be earlier?" using only observational records.
- *Data from Layer 2, query in Layer 2.* "Use the findings of previous experiments to predict the outcome of a new one" (transportability, generalization of experiments).
- *Data from Layers 1 and 2, query in Layer 3.* "Would I have won had I picked door #2 instead of door #1?"

Later lectures prove that, in general, lower layers *underdetermine* higher ones: without further assumptions, Layer 1 data cannot answer Layer 2 questions, and Layer 2 data cannot answer Layer 3 questions.

1.1.2.3 Two motivations

- **Scientific perspective:** understanding nature and human beings, using empirical data, logic and mathematics, and the scientific method.
- **Pragmatic perspective** (economics, medicine, engineering): building truly intelligent systems, and designing and implementing policies aimed at social, economic or political goals. Policies are interventions, so they live in Layer 2.

1.1.3 Correlation versus causation

Motivation. Show with real-world data that strong association is compatible with no causal relation at all.

Several famous spurious correlations:

- The number of active Facebook users and the yield of 10-year Greek government bonds (2005–2011) rise together (BusinessWeek, "Correlation or causation?", Dec. 2011).
- Fresh lemons imported to the USA from Mexico and total US highway fatality rate, 1996–2000, have $R^2 = 0.97$ (negative slope).
- Organic food sales and diagnosed autism prevalence, 1998–2007, have $r = 0.9971$ ($p < 0.0001$).
- The divorce rate in Maine correlates with the per-capita consumption of margarine (2000–2009).

Each pair is driven by common time trends or coincidence rather than by a mechanism linking them. The xkcd cartoon (no. 552) jokes: "I used to think correlation implied causation. Then I took a statistics class. Now I don't." "Sounds like the class helped." "Well, maybe." The real question is the converse one: *if correlation does not imply causation, then what does?* The answer developed in this course is: correlation *plus* causal assumptions, encoded in a model.

1.1.4 Example 1: Simpson's paradox in a 2×2 table

Motivation. Demonstrate that the same data support opposite conclusions depending on the causal story, so that no purely statistical rule decides which table to use.

Notation: X = took the drug, $\neg X$ = did not; Y = survived (recovered), $\neg Y$ = did not; F = female, $\neg F$ = male (Simpson 1951; Pearl 2000, Ch. 6).

All	Y	$\neg Y$	total	rate
drug (X)	20	20	40	50%
no drug ($\neg X$)	16	24	40	40%
Σ	36	44	80	

Male ($\neg F$)	Y	$\neg Y$	total	rate
drug	18	12	30	60%
no drug	7	3	10	70%
Σ	25	15	40	

Female (F)	Y	$\neg Y$	total	rate
drug	2	8	10	20%
no drug	9	21	30	30%
Σ	11	29	40	

Within each sex the drug is worse,

$$P(Y \mid F, X) = 0.2 < 0.3 = P(Y \mid F, \neg X),$$
$$P(Y \mid \neg F, X) = 0.6 < 0.7 = P(Y \mid \neg F, \neg X),$$

yet in the aggregate it is better: $P(Y \mid X) = 0.5 > 0.4 = P(Y \mid \neg X)$.

Why this is arithmetically possible. By the law of total probability,

$$P(Y \mid X) = P(Y \mid X, \neg F)P(\neg F \mid X) + P(Y \mid X, F)P(F \mid X).$$

Among drug takers $P(F \mid X) = 10/40 = 0.25$, while among non-takers $P(F \mid \neg X) = 30/40 = 0.75$. Hence

$$P(Y \mid X) = 0.75 \cdot 0.6 + 0.25 \cdot 0.2 = 0.50,$$
$$P(Y \mid \neg X) = 0.25 \cdot 0.7 + 0.75 \cdot 0.3 = 0.40.$$

The aggregate rates average the stratum rates with *different weights*; men, who recover more often regardless of treatment, are over-represented among drug takers. The reversal is not a paradox of probability, only of intuition.

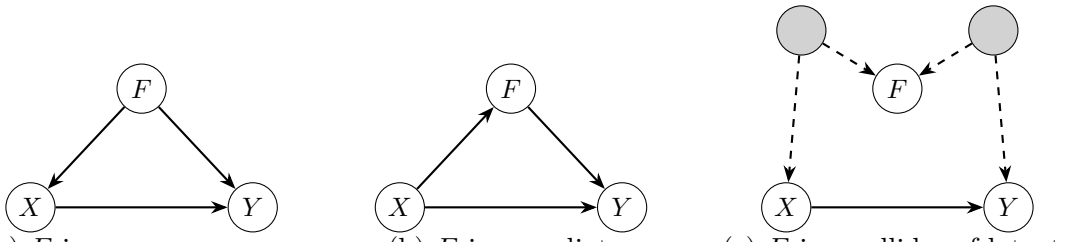
Which table should the physician consult? If sex affects both who takes the drug and recovery, the stratified table should be used, and the population-level effect is obtained by weighting the strata by the *population* distribution of sex, $P(F) = P(\neg F) = 40/80 = 1/2$:

$$\textstyle\sum_f P(Y \mid X, f)P(f) = \frac{1}{2}(0.6) + \frac{1}{2}(0.2) = 0.40,$$
$$\textstyle\sum_f P(Y \mid \neg X, f)P(f) = \frac{1}{2}(0.7) + \frac{1}{2}(0.3) = 0.50,$$

so the drug lowers recovery by 10 percentage points. (This formula is the back-door adjustment of Chapter 2.)

1.1.4.1 Would the result be different with a different story?

Keep the numbers but change the causal story behind them. Three possible graphs:



(a) F is a common cause (b) F is a mediator (c) F is a collider of latents

- **(a)** $F \to X$, $F \to Y$, $X \to Y$. F (sex) confounds the treatment–outcome relation. Consult the stratified tables; the drug is harmful.
- **(b)** $X \to F \to Y$, $X \to Y$. Suppose the third variable were not sex but something the drug changes (e.g. blood pressure). Stratifying on it would block part of the drug's effect; the aggregate table gives the total effect and the drug is beneficial.
- **(c)** $X \leftarrow U_1 \to F \leftarrow U_2 \to Y$, $X \to Y$ with U_1, U_2 unobserved. F is a collider; conditioning on it *creates* a spurious association between X and Y. The aggregate table is correct.

The data are identical in the three scenarios, yet the right answer differs. The decision therefore depends on knowledge outside the data, the causal structure, which is exactly what causal models represent.

1.1.5 Example 2: a "trivial" data analysis

Motivation. Show that a perfect fit does not reveal the direction or meaning of a dependence.

Firefighters (X)	Size of fire (Y)
10	100
40	400
60	600
200	2000

The points lie exactly on $Y = 10X$, so the regression $Y = \beta X + \beta_0$ fits perfectly with $\beta = 10$, $\beta_0 = 0$. X and Y are perfectly correlated, but does X cause Y? Read causally, $\beta = 10$ would say that each additional firefighter adds ten units of fire, which is absurd: the size of the fire determines how many firefighters are dispatched ($Y \to X$). *After all, what is the meaning of β?* A regression coefficient is a property of the joint distribution; whether it measures the effect of changing X depends on the causal mechanism, which the fit is silent about.

1.1.6 What is going on here?

Both examples ask for something that is not in the joint distribution: the effect of an action. Answering such questions requires (i) a formal language for causal assumptions, (ii) a semantics connecting assumptions with data and queries, and (iii) inference rules deciding when and how a causal query is computable from available data. Chapters 1–3 build exactly this machinery: structural causal models, causal graphs, identification, the back-door and adjustment criteria, do-calculus, and algorithmic identification.

1.2 Causal models and causal graphs

Motivation. The central question is what object can play the role of an "oracle" for causal questions, given that the observational distribution alone cannot determine the right decision, and what that object implies about the data we do observe.

1.2.1 Reviews: notation, probability, algebra

Motivation. Fix notation and the elementary manipulations used throughout the course.

1.2.1.1 Notation

- A random variable is denoted by an uppercase letter, e.g. Z, with state space $\mathfrak{X}_Z$ (rarely written explicitly). A lowercase letter denotes a value, $z \in \mathfrak{X}_Z$.
- Bold letters denote sets of variables or values, e.g. $\mathbf{V}$ and $\mathbf{v}$. Calligraphic letters denote complex objects such as graphs $\mathcal{G}$ and models $\mathcal{M}$.
- $\forall$ (for all), $\exists$ (exists); $\cup, \cap$; $\subseteq, \subset$ (note that $\subset$ is ambiguous between proper and non-proper subset); $\vee$ (or), $\wedge$ (and), $\oplus$ (xor); $\sum$, $\prod$; $\langle \cdots \rangle$ tuples, $\{\cdots\}$ sets, $\mid$ "given".
- "iff" = if and only if. $p \Rightarrow q$: if p then q; p is *sufficient* for q. Equivalently $\neg q \Rightarrow \neg p$: q is *necessary* for p.

1.2.1.2 Probability

Consider $P(A, B, C, D, E, F)$; write $P(A = a, B = b)$ as $P(a, b)$. Variables are assumed discrete without loss of generality.

Marginalization.

$$\sum_{c,e} P(a, b, c, d, e, f) = \sum_{(c,e) \in \mathfrak{X}_C \times \mathfrak{X}_E} P(a, b, c, d, e, f) = P(a, b, d, f).$$

Conditioning.

$$P(a, b, c \mid d, e) = \frac{P(a, b, c, d, e)}{P(d, e)} = \frac{\sum_f P(a, b, c, d, e, f)}{\sum_{a,b,c} \sum_f P(a, b, c, d, e, f)}.$$

Chain rule. Applying the definition of conditional probability recursively, in *any* order:

$$P(a, b, d, e) = P(a \mid b, d, e) P(b \mid d, e) P(d \mid e) P(e) = P(b \mid a, d, e) P(d \mid a, e) P(e \mid a) P(a).$$

Axioms. $0 \leq P(\cdot) \leq 1$ and $\sum_{a,b,c} P(a, b, c) = 1$. Often strictly positive distributions are assumed so that every conditional is defined.

1.2.1.3 Sums of products

Factors that do not depend on a summation index can be moved outside that sum:

$$\sum_{a,c} f(a, b) g(b, c) h(c, d) = \sum_c h(c, d) g(b, c) \Big(\sum_a f(a, b) \Big) = \sum_a f(a, b) \Big(\sum_c g(b, c) h(c, d) \Big).$$

If $f(a, b) = P(a \mid b)$, then $\sum_a P(a \mid b) = 1$ and

$$\sum_a P(a \mid b) \underbrace{\sum_c g(b, c) h(c, d)}_{\text{irrelevant to } a} = \underbrace{\Big(\sum_a P(a \mid b) \Big)}_{=1} \sum_c g(b, c) h(c, d) = \sum_c g(b, c) h(c, d).$$

This "sum out a conditional of a leaf" step is used constantly in identification proofs.

1.2.2 Why data alone is not enough: a prima facie example

Motivation. Construct a situation in which every learning algorithm, trained on observational data, gives the wrong policy recommendation.

1.2.2.1 The story

A city suffers from a contagious disease, and a drug is *believed* to help. Unknown to physicians:

- People with good living conditions (rich) always survive.
- Some people carry a gene that naturally fights the disease; they need no treatment. If treated, however, they develop an allergic reaction that is fatal under poor living conditions.

Being rich and having the gene are independent, $P(\text{rich}) = P(\text{poor}) = \frac{1}{2}$, and half the poor have the gene. The population Π splits into three groups:

$$\Pi = \text{rich} \ \cup \ \text{poor}_1 \ \cup \ \text{poor}_2,$$

$$\begin{cases} \text{rich}: & \text{alive anyway}, \\ \text{poor}_1 \ (\text{no gene}): & \text{die anyway}, \\ \text{poor}_2 \ (\text{gene}): & \text{die iff they take the drug}, \end{cases}$$

with $P(\text{poor}_1) = P(\text{poor}_2) = \frac{1}{4}$. (*Prima facie* = at first sight.)

1.2.2.2 Current protocol

Because the drug is expensive, physicians prescribe it to every rich person and only to them. Observed outcomes: all treated (rich) survive; among the untreated (poor) poor_2 survive and poor_1 die. Hence

$$P(\text{alive} \mid \text{drug}) = 1, \qquad P(\text{alive} \mid \neg\text{drug}) = 0.5.$$

Should the city give the drug to everyone for free?

1.2.2.3 Hypothetical protocols

Group (mass)	Current (drug iff rich)	Drug for all	Drug for none
rich (1/2)	alive	alive	alive
poor_1 (1/4)	dead	dead	dead
poor_2 (1/4)	alive	dead	alive
$P(\text{alive})$	3/4	1/2	3/4

Giving the drug to everyone *reduces* survival from 3/4 to 1/2; the drug never helps anyone. The apparent benefit is produced entirely by the fact that only the rich, who survive anyway, are treated.

1.2.2.4 What can a learning system see?

A data set has rows (patient, R, D, A), e.g. (1, yes, yes, yes), (2, no, no, yes), (3, no, no, no), (4, yes, yes, yes),... The best any algorithm (decision tree, SVM, DNN) can do is to learn the joint distribution

R	D	A	$P(r,d,a)$
0	0	0	1/4
0	0	1	1/4
1	1	1	1/2
others			0

What remains unseen are the exogenous factors: the probability of having the allergic gene, $P(\text{gene})$; the other factors that affect wealth; and how the genetic factor interacts with wealth, the drug and the reaction.

What is missing? Consider a second world in which the drug is simply inert, $A \leftarrow R \vee U_G$. Since $D = R$ in the data, this world produces exactly the same table, but there "drug for all" gives survival 3/4. Two realities with different answers generate identical data, so no algorithm using only $P(r,d,a)$ can be correct in both. The data must be supplemented with a model of how the data are generated.

1.2.2.5 From statistical to causal analysis

- **Traditional statistics/ML:** data $\rightarrow$ joint distribution $P \rightarrow$ query $Q(P)$, an aspect of P, e.g. will customers who bought A also buy B, $Q = P(B \mid A)$.
- **Causal analysis:** data $\rightarrow P \xrightarrow{\text{change}} P' \rightarrow Q(P')$, e.g. $P'(\text{sales})$ if we double the price, $P'(\text{cancer})$ if smoking is banned. How does the factual P change into the hypothetical P'? P is tied to data; P' is never observed. A new formalism must represent both.
- **SCM paradigm:** data $\leftarrow P \leftarrow \mathcal{M}$ (data-generating model) $\rightarrow Q(\mathcal{M})$. $\mathcal{M}$ is an invariant strategy (mechanism, recipe, law, protocol) by which Nature assigns values to the variables. *P is a model of data; $\mathcal{M}$ is a model of reality.*

In the big picture of Section 1.1, knowledge of the real world enters the causal model, and causal inference draws new knowledge about not-yet-realized alternative realities. (This picture should be taken with a grain of salt.)

1.2.2.6 Modeling reality in the example

Observed (endogenous) variables $\mathbf{V}$: R (1 = rich), D (1 = took the drug), A (1 = alive). Unobserved (exogenous) variables $\mathbf{U}$: U_G (1 = has the genetic factor), U_R (other factors affecting wealth). Mechanisms:

$$R \leftarrow U_R, \qquad D \leftarrow R, \qquad A \leftarrow R \vee (U_G \wedge \neg D),$$

with $P(U_G = 1) = P(U_R = 1) = \frac{1}{2}$ and $P(\mathbf{U}) = P(U_G)P(U_R)$. This is a fully specified model of reality: it implies both P and every P'. It is our new oracle, the *structural causal model.*

1.2.3 Structural causal models

Motivation. Define the object that represents reality and derive its three basic properties.

Definition (Structural causal model; Pearl 2000)

An SCM $\mathcal{M}$ is a 4-tuple $\langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$ where

- $\mathbf{V} = \{V_1, \ldots, V_n\}$ are endogenous variables;
- $\mathbf{U}$ is a set of exogenous variables;
- $\mathcal{F} = \{f_1, \ldots, f_n\}$ are functions determining $\mathbf{V}$, $v_i \leftarrow f_i(\mathbf{pa}_i, \mathbf{u}_i)$, with $Pa_i \subseteq \mathbf{V} \setminus \{V_i\}$

and $\mathbf{U}_i \subseteq \mathbf{U}$;

- $P(\mathbf{U})$ is a distribution over $\mathbf{U}$.

The arrow "$\leftarrow$" stresses that f_i is an assignment (a mechanism), not a symmetric equation. Axiomatic characterizations of the resulting counterfactual logic are given by Galles and Pearl (1998) and Halpern (1998). Abstractly: the exogenous $\mathbf{U}$ carry all randomness; $\mathcal{F}$ deterministically transforms a unit $\mathbf{u}$ into its observable profile.

1.2.3.1 Property 1: an SCM induces the observational distribution

$\mathcal{F}$ is a map $\mathbf{u} \mapsto (v_1, \ldots, v_n)$. When $\mathbf{U}$ is random, so is $\mathbf{V}$. The induced $P(\mathbf{v})$ is Layer 1 of the PCH, the observational (passive) distribution. Each event, person or observation corresponds to an instantiation $\mathbf{U} = \mathbf{u}$.

Definition (Induced distribution)

$\mathcal{M}$ defines $P^{\mathcal{M}}(\mathbf{V})$ by, for each $\mathbf{Y} \subseteq \mathbf{V}$,

$$P^{\mathcal{M}}(\mathbf{y}) = \sum_{\{\mathbf{u}\,:\,\mathbf{Y}(\mathbf{u})=\mathbf{y}\}} P(\mathbf{u}),$$

where $\mathbf{Y}(\mathbf{u})$, the *potential response*, is the value of $\mathbf{Y}$ obtained by solving $\mathcal{F}$ at $\mathbf{u}$.

Example 1.1

In the drug example each citizen falls into one of four groups:

$P(\mathbf{u})$	(U_R, U_G)	$\xrightarrow{\mathcal{F}} (R, D, A)$	$P(\mathbf{v})$
1/4	(1,1)	(1,1,1)	1/2
1/4	(1,0)	(1,1,1)	
1/4	(0,1)	(0,0,1)	1/4
1/4	(0,0)	(0,0,0)	1/4

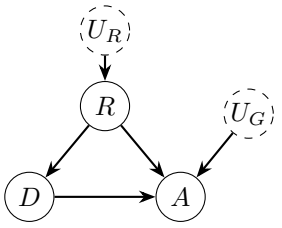
1.2.3.2 Property 2: an SCM induces a causal diagram

Definition (Causal diagram; Bareinboim et al. 2020, Def. 13)

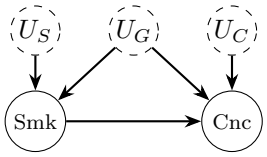
Given $\mathcal{M} = \langle \mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}) \rangle$, the causal diagram $\mathcal{G}$ is constructed by

1. adding a vertex for every $V_i \in \mathbf{V}$;
2. adding a directed edge $V_j \to V_i$ whenever V_j is an argument of f_i;
3. adding a bidirected edge $V_j \leftrightarrow V_i$ whenever $\mathbf{U}_i$ and $\mathbf{U}_j$ are correlated or f_i, f_j share some $U \in \mathbf{U}$ as an argument.

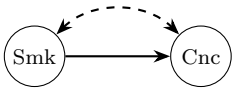
More precisely, for every set $\mathbf{W}$ of vertices with no bidirected edges among them, $\{\mathbf{U}_W\}_{W \in \mathbf{W}}$ are jointly independent. If the directed part is acyclic (assumed throughout) the diagram is a DAG with possible bidirected edges (an ADMG). For instance, $V_i \leftarrow f_i(A, B, U)$ and $V_j \leftarrow f_j(C, U)$ give $A \to V_i \leftarrow B$, $C \to V_j$, and $V_i \leftrightarrow V_j$ because of the shared U.



drug example



explicit **U**



U implicit

Example 1.2 (Smoking and cancer)

$\mathbf{V} = \{\text{Smoking}, \text{Cancer}\}$, $\mathbf{U} = \{U_S, U_C, U_G\}$, Smoking $\leftarrow f_S(U_S, U_G)$, Cancer $\leftarrow f_C(\text{Smoking}, U_C, U_G)$. The shared genetic factor U_G yields Smoking $\leftrightarrow$ Cancer.

Remark 1. The mapping goes one way only: a graph is compatible with infinitely many SCMs sharing the same function signatures and exogenous structure.

Remark 2. This is central: in practice researchers often know the *scope* of the mechanisms (which variables enter which function) but not their form. Causal inference asks what can be concluded from the graph plus data alone.

Convention. Unobserved variables are left implicit; a shared or correlated U is drawn as a dashed bidirected edge.

1.2.3.3 Graph terminology

A graph $\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$ consists of vertices (nodes) and edges. For $V \in \mathbf{V}$:

- parents $Pa(V) = \{W : W \to V \in \mathcal{G}\}$; children $Ch(V) = \{W : V \to W \in \mathcal{G}\}$;
- ancestors $An(V) = \{W : W \to \cdots \to V \in \mathcal{G}\}$; descendants $De(V) = \{W : V \to \cdots \to W \in \mathcal{G}\}$; these include V itself (zero-length paths);
- siblings (spouses through confounding) $\{W : W \leftrightarrow V \in \mathcal{G}\}$;
- a *path* is a sequence of edges joining a sequence of distinct vertices; a *directed path* from W to V has all arrowheads pointing toward V.

1.2.3.4 Property 3: an SCM induces Pearl's Causal Hierarchy

An SCM $\mathcal{M}$ answers queries of all three layers: association $P(y \mid x)$, intervention $P(y \mid \text{do}(x), c)$ and counterfactual $P(y_x \mid x', y')$. This lecture concentrates on Layer 1; Layers 2 and 3 are formalized in Chapter 2.

1.2.4 The first layer: what the diagram encodes about $P(\mathbf{v})$

Motivation. Derive the factorization of the observational distribution implied by an SCM and its diagram.

$\mathcal{M}$ induces

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{V} = \mathbf{v}, \mathbf{U} = \mathbf{u}) = \sum_{\mathbf{u}} P\big(\forall_{V_i \in \mathbf{V}}\, f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i,\ \mathbf{U} = \mathbf{u}\big),$$

where $\mathbf{pa}_i, \mathbf{u}_i$ are the observed and unobserved arguments of f_i, i.e. the parents of V_i in $\mathcal{G}$. In the examples,

$$P(r, d, a) = \sum_{u_R, u_G} P\big(\underbrace{r, d, a}_{\text{endogenous}}, \underbrace{u_R, u_G}_{\text{exogenous}}\big),$$

$$P(s,c) = \sum_{u_S,u_G,u_C} P(s,c,u_S,u_G,u_C).$$

This is the observational (passive, non-experimental) distribution.

Topological order. Because $\mathcal{G}$ is acyclic, there is an order $\prec$ on $\mathbf{V}$ in which every variable comes after its parents, $Pa_i \prec V_i$; e.g. $R \prec D \prec A$ and $S \prec C$.

Factorization. Apply the chain rule along a topological order:

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}).$$

Each V_i is fully determined by its observed and unobserved parents, and $Pa_i \cup \mathbf{U}_i \subseteq \{V_1, \ldots, V_{i-1}\} \cup \mathbf{U}$, so

$$P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P\big(f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}\big) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

Therefore

$$\boxed{P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).}$$

Examples:

$$P(r,d,a) = \sum_{u_R,u_G} P(u_R,u_G)P(r \mid u_R)P(d \mid r)P(a \mid r,d,u_G),$$
$$P(s,c) = \sum_{u_S,u_G,u_C} P(u_S,u_G,u_C)P(s \mid u_G,u_S)P(c \mid s,u_G,u_C).$$

Here $P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) \in \{0,1\}$ because the mechanisms are deterministic.

1.2.5 Conditional independence and the Markov condition

Motivation. Translate the structure of $\mathcal{G}$ into testable constraints on $P(\mathbf{v})$.

1.2.5.1 Conditional independence

X and Y are independent, $X \perp\!\!\!\perp Y$, if learning $X = x$ does not change the belief in $Y = y$:

$$X \perp\!\!\!\perp Y \iff P(y \mid x) = P(y) \;\; \forall x, y \iff P(x,y) = P(x)P(y).$$

X and Y are conditionally independent given Z if, once $Z = z$ is known, X is uninformative about Y:

$$(X \perp\!\!\!\perp Y \mid Z) \iff P(x \mid z, y) = P(x \mid z) \quad \forall x, y, z \text{ with } P(y,z) > 0.$$

These are constraints on $P(\mathbf{V})$. The guiding intuition: *lack of functional dependence leads to probabilistic independence.*

1.2.5.2 Local Markov condition

In any topological order, $P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$. What is the largest set that can precede V_i in some topological order? Any W that is not reachable by a directed path $V_i \to \cdots \to W$, i.e. all *non-descendants* Nd_i of V_i. Therefore

$$P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = P(v_i \mid \mathbf{nd}_i, \mathbf{u}),$$

which is the statement

$$V_i \perp\!\!\!\perp (Nd_i \setminus Pa_i), \; (\mathbf{U} \setminus \mathbf{U}_i) \;\mid\; Pa_i, \mathbf{U}_i.$$

1.2.5.3 Markovian models and the Bayesian factorization

If $U_1, \ldots, U_n$ are jointly independent (no bidirected edges), the model is *Markovian*. Then

$$\begin{aligned} P(\mathbf{v}) &= \sum_{\mathbf{u}} P(\mathbf{u}) \prod_i P(v_i \mid \mathbf{pa}_i, u_i) \\ &= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i) && (U_i \text{ independent}) \\ &= \sum_{\mathbf{u}} \prod_i P(v_i \mid \mathbf{pa}_i, u_i) P(u_i \mid \mathbf{pa}_i) && (U_i \perp\!\!\!\perp Pa_i) \\ &= \sum_{\mathbf{u}} \prod_i P(v_i, u_i \mid \mathbf{pa}_i) \\ &= \prod_i \sum_{u_i} P(v_i, u_i \mid \mathbf{pa}_i) = \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i). \end{aligned}$$

The penultimate step uses that the i-th factor depends only on u_i, so the sum over $\mathbf{u}$ distributes over the product. $U_i \perp\!\!\!\perp Pa_i$ holds because Pa_i are functions of exogenous variables of non-descendants, which are independent of U_i in a Markovian model.

Summary. Markovian SCM $\Rightarrow$ $P(\mathbf{v}) = \prod_i P(v_i \mid \mathbf{pa}_i)$ (Bayesian factorization) $\Leftrightarrow$ local Markov condition $V_i \perp\!\!\!\perp Nd_i \setminus Pa_i \mid Pa_i$.

1.2.6 Reading independences from the graph: d-separation

Motivation. Connect the unobserved mechanisms with observable independences through three elementary structures.

Each example below uses binary variables and explicit mechanisms.

1.2.6.1 Chains

$X \to Z \to Y$: good study habits $X \leftarrow U_X$, a high Suneung score $Z \leftarrow X \vee \neg U_Z$ (U_Z: difficulty), college admission $Y \leftarrow Z \wedge U_Y$ (U_Y: favorable factors). Are X and Y independent given Z? **Yes.** Knowing $Z = 1$, learning about study habits does not change the admission probability:

$$P(x, y \mid z) = \frac{P(x, y, z)}{P(z)} = \frac{P(y \mid z)P(z \mid x)P(x)}{P(z)} = \frac{P(x, z)}{P(z)} P(y \mid z) = P(x \mid z)P(y \mid z).$$

Observing Z blocks the influence from X to Y. (Marginally, X and Y are dependent.)

1.2.6.2 Common causes (forks)

$X \leftarrow Z \to Y$: rain $Z \leftarrow U_Z$ (meteorological conditions), traffic $X \leftarrow Z \vee \neg U_X$ (road design), umbrella $Y \leftarrow Z \wedge U_Y$. Are X, Y independent? **No**: seeing an umbrella raises the probability of rain, which raises the probability of bad traffic; $\exists x, y$ with $P(x, y) \neq P(x)P(y)$. Information flows from Y up to Z and down to X. Given Z? **Yes**: once we know it is raining, umbrellas tell us nothing about traffic,

$$P(x, y \mid z) = \frac{P(x \mid z)P(y \mid z)P(z)}{P(z)} = P(x \mid z)P(y \mid z).$$

1.2.6.3 Common effects (colliders)

$X \to Z \leftarrow Y$: having a cold $X \leftarrow U_X$, holiday $Y \leftarrow U_Y$, going to class $Z \leftarrow \neg Y \wedge (\neg X \wedge U_Z)$ (U_Z: motivation). Are X, Y independent? **Yes**:

$$P(x,y) = \sum_z P(x)P(y)P(z \mid x,y) = P(x)P(y)\sum_z P(z \mid x,y) = P(x)P(y).$$

Influence from X reaches Z but does not "go up" to Y. Given Z? **No!** If a student did not go to class ($Z = 0$) and it is not a holiday ($Y = 0$), a cold becomes more likely; $\exists x, y, z$ with $P(x, y \mid z) \neq P(x \mid z)P(y \mid z)$ ("explaining away"). This is the opposite of the previous two cases.

Given a descendant W of the collider (signing the attendance sheet, $W \leftarrow Z \wedge U_W$)? **No, again.** Observing $W = 0$ makes absence ($Z = 0$) more likely, which again couples X and Y. *Watch out for descendants of colliders.*

1.2.6.4 Triplet summary

Triplet	Active (dependence flows) when	Inactive when
chain $X \to Z \to Y$	Z not observed	Z observed
fork $X \leftarrow Z \to Y$	Z not observed	Z observed
collider $X \to Z \leftarrow Y$	Z or a descendant of Z observed	neither Z nor its descendants observed

1.2.6.5 d-separation

Definition (d-separation)

To decide whether $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ (with $X, Y \notin \mathbf{Z}$):

1. look at every path between X and Y;
2. a path is *active* given $\mathbf{Z}$ if every consecutive triplet on it is active;
3. if any path is active, X and Y are d-connected (not guaranteed independent); otherwise they are d-separated, written $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}}$.

Trivially, if $X = Y$ or X and Y are adjacent, they are dependent.

Soundness: if $(X \perp_d Y \mid \mathbf{Z})_{\mathcal{G}}$ then $X \perp\!\!\!\perp Y \mid \mathbf{Z}$ in every distribution induced by an SCM compatible with $\mathcal{G}$ (the global Markov property, equivalent to the local one for DAGs).

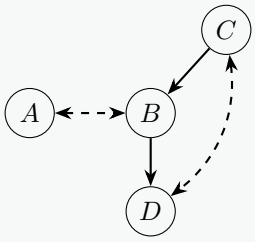
Example 1.3 (Sprinkler network)

Season $\to$ Sprinkler, Season $\to$ Rain, Sprinkler $\to$ Wet, Rain $\to$ Wet, Wet $\to$ Slippery.

1. Wet $\perp\!\!\!\perp$ Sprinkler? **No**, they are adjacent.
2. Wet $\perp\!\!\!\perp$ Season $\mid$ Sprinkler? **No**: Season $\to$ Rain $\to$ Wet is open.
3. Rain $\perp\!\!\!\perp$ Slippery $\mid$ Wet? **Yes**: the only path is the chain through Wet.
4. Season $\perp\!\!\!\perp$ Wet $\mid$ Sprinkler, Rain? **Yes**: both directed paths are blocked at observed non-colliders.
5. Sprinkler $\perp\!\!\!\perp$ Rain $\mid$ Season, Wet? **No**: Wet is an observed collider on Sprinkler $\to$ Wet $\leftarrow$ Rain.

Example 1.4 (Food for thought [Supplement])

Graph: $A \leftrightarrow B$, $C \to B$, $B \to D$, $C \leftrightarrow D$.



- $A \perp\!\!\!\perp D$? **No**: on $A \leftrightarrow B \to D$, B is an unobserved non-collider.
- $A \perp\!\!\!\perp C$? **Yes**: $A \leftrightarrow B \leftarrow C$ has a collider at B; $A \leftrightarrow B \to D \leftrightarrow C$ has a collider at D.
- $A \perp\!\!\!\perp C \mid D$? **No**: B is an ancestor of the observed D, so the collider at B is opened.
- $D \perp\!\!\!\perp C \mid B$? **No**: $D \leftrightarrow C$ are adjacent (confounded).

1.2.7 Implementation: graph algorithms in Python

Motivation. Turn the definitions into code: topological ordering and d-separation.

1.2.7.1 Topological order

Idea: a DAG has a source node (no parents); removing a node and its edges leaves a DAG. Find a source, then recursively order the rest.

```
def topological_order(G):
    for V in G.Vs:                  # for every node
        if not G.Pa(V):             # if there is no parent for V
            return [V] + topological_order(G - V)
    return []
```

This naive version is quadratic and silently returns a partial order on cyclic graphs. Tracking in-degrees (Kahn's algorithm) is efficient and detects cycles:

```
def topological_order(G):
    order = deque()
    in_degrees = {V: len(G.Pa(V)) for V in G.Vs}
    while in_degrees:
        for v, v_deg in in_degrees.items():
            if v_deg == 0:
                break
        else:
            raise ValueError('cyclic')
        order.append(v)
        del in_degrees[v]
        for c in G.Ch(v):
            in_degrees[c] -= 1
    return order
```

The `for`–`else` clause runs only if no zero-in-degree node was found, which happens iff the remaining graph contains a cycle.

1.2.7.2 d-separation via reachability (Markovian graphs)

X and Y are d-connected given $\mathbf{Z}$ iff there is a path along which every triplet $A - B - C$ satisfies:

$$\begin{aligned} A \to B \to C \quad & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \leftarrow C \quad & \text{only if } B \notin \mathbf{Z}, \\ A \leftarrow B \to C \quad & \text{only if } B \notin \mathbf{Z}, \\ A \to B \leftarrow C \quad & \text{only if } B \in An(\mathbf{Z}). \end{aligned}$$

So we walk from B to C only if the condition holds, using BFS/DFS over states (direction of arrival, B): arrived along an edge pointing *into* B ($\to B$) or *out of* B ($\leftarrow B$).

```
def d_separation(G, X, Y, Zs):
    q = visited_deque({('->', X), ('<-', X)})
    while q:
        direction, B = q.pop()
        if B == Y:
            return False
        if direction == '->':          # arrived with arrowhead into B
            if B in G.An(Zs):          # collider: open iff B in An(Z)
                q.extend({('<-', C) for C in G.Pa(B)})
            if B not in Zs:            # chain continues
                q.extend({('->', C) for C in G.Ch(B)})
        else:                          # arrived via tail of B (B -> previous)
            if B not in Zs:
                q.extend({('->', C) for C in G.Ch(B)})
                q.extend({('<-', C) for C in G.Pa(B)})
    return True
```

`visited_deque` is a deque that ignores states already visited, so each (direction, node) pair is processed once and the algorithm is linear in the size of the graph. The starting states allow leaving X in either direction.